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Vulnerability Assessment Report

Executive Summary

Target Host: Metasploitable2 (10.0.2.3)
Assessment Date: August 3, 2026
Scan Duration: 11:53 AM – 12:01 PM (8 minutes)
Scanner: Tenable Nessus Essentials (Local Scanner)
Scan Policy: Basic Network Scan
Severity Scoring Standard: CVSS v3.0

A comprehensive vulnerability assessment was performed on the target host **10.0.2.3** using **Tenable Nessus Essentials**. The objective of the assessment was to identify security weaknesses that could be exploited by attackers and to provide recommendations for remediation.

The scan detected **40 vulnerability instances** across multiple severity levels. The findings include **critical remote code execution vulnerabilities, outdated software versions, CGI-related information disclosure issues**, and several configuration weaknesses. These vulnerabilities present varying levels of risk to the confidentiality, integrity, and availability of the target system.

Severity Breakdown

Severity Level	CVSS v3.0 Range	Findings	Percentage
Critical	9.0 – 10.0	2	5.0%
High	7.0 – 8.9	0	0.0%
Medium	4.0 – 6.9	4	10.0%

Low	0.1 – 3.9	2	5.0%
Mixed / Multiple	N/A	22	55.0%
Informational	0.0	10	25.0%
Total	—	40	100%

Detailed Vulnerability Analysis

1. Critical Severity Findings

1.1 High-Risk Service Vulnerability

- **CVSS Score:** 10.0 (Critical)
- **Vulnerability Family:** General
- **Occurrences:** 1

Description

The assessment identified an exposed service containing an unauthenticated remote execution vector or backdoor. An attacker can exploit this vulnerability without providing valid credentials.

Risk Impact

Successful exploitation would allow a remote attacker to obtain **root-level access** to the operating system, resulting in complete compromise of:

- System confidentiality
- Data integrity
- Service availability

Recommended Remediation

- Immediately disable or remove the vulnerable legacy service.
- Upgrade to a supported and patched version.

- Restrict network exposure using firewall rules or access control lists (ACLs).
 - Continuously monitor for unauthorized access attempts.
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1.2 Unauthenticated Service Exploitation

- **CVSS Score:** 9.8 (Critical)
- **Vulnerability Family:** Service Detection
- **Occurrences:** 1

Description

A vulnerable software version was detected that permits remote code execution or authentication bypass without requiring user credentials.

Risk Impact

An attacker could remotely execute arbitrary code, leading to full system compromise and potential lateral movement within the network.

Recommended Remediation

- Upgrade the affected software to the latest supported version.
 - Apply all available vendor security patches.
 - Disable unnecessary services until remediation is completed.
 - Perform validation scans after patching.
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2. Medium Severity Findings

2.1 Outdated Service Detection

- **CVSS Score:** 6.5
- **Vulnerability Family:** Service Detection

Description

Several services are running outdated software versions that expose system information and contain publicly known vulnerabilities.

Potential Impact

- Increased attack surface
- Easier fingerprinting by attackers
- Higher likelihood of exploitation through known vulnerabilities

Recommended Remediation

- Update all outdated services.
 - Remove unsupported software.
 - Establish a regular patch management process.
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2.2 Miscellaneous Service Weakness

- **CVSS Score:** 5.9
- **Vulnerability Family:** Miscellaneous

Description

Configuration weaknesses were identified that may enable privilege escalation or unauthorized system access.

Potential Impact

Attackers could exploit weak configurations to gain elevated privileges or bypass security controls.

Recommended Remediation

- Review service configurations.
 - Apply the principle of least privilege.
 - Remove unnecessary permissions and services.
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2.3 CGI Information Disclosure

- **CVSS Score:** 5.0

- **Vulnerability Family:** CGI Abuses
- **Occurrences:** 2

Description

The web server exposes sensitive information through:

- Accessible `.svn/entries` repository metadata
- CGI `printenv` output revealing server environment variables

Potential Impact

Information disclosure may assist attackers in gathering intelligence for subsequent attacks.

Recommended Remediation

- Remove exposed Subversion metadata.
- Disable unnecessary CGI scripts.
- Restrict access to administrative resources.
- Configure the web server to prevent sensitive file disclosure.

3. Low and Mixed Severity Findings

Low Severity Findings

Total Findings: 2

The assessment identified minor security issues including:

- ICMP timestamp responses
- General information disclosure

Although these vulnerabilities have limited direct impact, they can aid attackers during reconnaissance activities.

Recommended Remediation

- Disable ICMP timestamp responses where operationally feasible.

- Minimize unnecessary information disclosure from network services.
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Mixed Severity Findings

Total Findings: 22

These findings span multiple vulnerability families, including:

- General service checks
- Web server configuration issues
- HTTP multi-issue scripts
- WebDAV detection
- Miscellaneous service weaknesses

While individually these findings may present moderate or low risk, collectively they increase the attack surface and should be addressed as part of routine system hardening.

Risk Summary

The assessment indicates that the target host contains **multiple critical vulnerabilities** capable of allowing complete system compromise through unauthenticated remote exploitation. Additionally, outdated software versions and insecure web server configurations significantly increase the likelihood of successful attacks.

Immediate remediation of critical vulnerabilities should be prioritized, followed by systematic patch management and security hardening to reduce overall organizational risk.

Strategic Remediation Roadmap

Immediate Actions (Within 24-48 Hours)

- Patch or isolate all critical vulnerabilities (CVSS 9.8-10.0).

- Disable vulnerable legacy services.
 - Restrict external access to exposed ports using firewall rules.
 - Verify remediation through follow-up vulnerability scans.
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Short-Term Actions (Within One Week)

- Upgrade outdated software and services.
 - Secure Apache and CGI configurations.
 - Remove exposed `.svn` directories and repository metadata.
 - Eliminate unnecessary information disclosure from web services.
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Long-Term Security Hardening

- Implement a host-based firewall (e.g., **iptables** or **UFW**) to restrict access to authorized networks.
 - Establish a regular vulnerability assessment and patch management schedule.
 - Enforce the principle of least privilege for services and user accounts.
 - Continuously monitor system logs and network activity for indicators of compromise.
 - Conduct periodic penetration testing to validate the effectiveness of implemented security controls.
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Conclusion

The vulnerability assessment identified several high-risk security issues requiring immediate attention, particularly the critical remote code execution vulnerabilities.

Addressing these findings promptly will significantly reduce the attack surface and improve the overall security posture of the target environment. Ongoing vulnerability management, timely patching, and secure configuration

practices are recommended to maintain long-term system security.

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